GraphEntry: from small BDV asymmetry to a global $C^{1,\gamma}$ normal graph

Abstract

This is the GraphEntry building block of the Plan 1 / Route A proof of sharp Faber–Krahn stability with exponent 1/2. It is a strictly quantitative, contradiction-free pipeline that turns a smallness hypothesis on the BDV asymmetry $\alpha(U)$ of a selected minimizer $U \subset B_R$ into a deterministic conclusion: ∂U is a global $C^{1,\gamma}$ normal graph over ∂B_1 with L^∞ norm controlled by $\alpha(U)^{1/(2N)}$ and $C^{1,\gamma}$ norm bounded by a fixed constant $M_1 = M_1(N, R)$. The Brasco–De Philippis–Velichkov (BDV) free boundary results (Lemmas 4.2, 4.9, 4.12, 4.16 and Theorem 4.18 of [1]) are invoked as quantitative black boxes; no compactness or sequential argument is used. The block accepts the output of SelectionPrinciple and feeds SchauderInterpolation; in particular the upgrade from $C^{1,\gamma}$ to small C^{2,γ_0} is *not* part of this note.

Contents

1 Standing data

1.1 Sets, functions, asymmetries

Throughout, $N \geq 2$ is the ambient dimension and $R > 1$ a fixed container radius. Let $U \subset B_R \subset \mathbb{R}^N$ be a measurable set with $|U| = |B_1| = \omega_N$, and $u \in H_0^1(B_R)$ its torsion potential with $\{u > 0\} = U$. The Hausdorff distance between two compact sets is

$$d_H(A, B) := \max\Big\{\sup_{a\in A} \operatorname{dist}(a, B),\ \sup_{b\in B} \operatorname{dist}(b, A)\Big\}.$$

The *Fraenkel asymmetry* of Ω is

$$\mathcal{A}(\Omega) := \inf_{x\in\mathbb{R}^N} \frac{|\Omega\Delta(B_1+x)|}{|B_1|},$$

and the *BDV (smooth) asymmetry* $\alpha(U)$ is the smooth penalised quantity defined in [1, §4].

1.2 BDV regularity package (black box)

Let $U = \{u > 0\}$ be a selected minimizer with selection parameter $\tau \le \tau_{\mathrm{reg}}(N,R)$. The BDV theorems give constants $c_d = c_d(N,R)$, $r_d = r_d(N,R)$, $L_u = L_u(N,R)$, $q_\pm = q_\pm(N,R)$, $L_q = L_q(N,R)$, all positive, such that:

(L4.9) **Two-sided density.** For every $x \in \partial U$ and $0 < r \le r_d$,

$$c_d r^N \le |U \cap B_r(x)| \le (1-c_d)r^N, \quad c_d r^N \le |B_r(x)\setminus U| \le (1-c_d)r^N.$$

(L4.12a) **Lipschitz.** $u \in C^{0,1}(\overline{B_R})$ with $\mathrm{Lip}(u) \le L_u$.

(L4.12b) **Nondegeneracy.** $\sup_{B_r(x)} u \ge c_d r$ for all $x \in \partial U$ and $0 < r \le r_d$.

(L4.16) **Bernoulli coefficient.** On the flat portion of ∂U, $q_u(x) := |\nabla u(x)|$ satisfies $q_- \le q_u \le q_+$ and $\mathrm{Lip}(q_u) \le L_q$.

(L4.2) **Asymmetry vs. symmetric difference.** $|U\Delta B_1|^2 \le C_1(N)\,\alpha(U)$, and $|\alpha(U_1)-\alpha(U_2)| \le C_2(R)|U_1\Delta U_2|$.

(T4.18) **Alt–Caffarelli flatness improvement.** There exist $\bar\mu = \bar\mu(N,R) \in (0,1)$, $\bar\rho = \bar\rho(N,R) > 0$, $\gamma = \gamma(N,R) \in (0,1)$, $C_{\mathrm{AC}} = C_{\mathrm{AC}}(N,R)$ such that if $u \in F(\bar\mu, 1, +\infty)$ on $B_{\bar\rho}(x_0) \subset B_R$, then $\partial U \cap B_{\bar\rho/2}(x_0)$ is a $C^{1,\gamma}$ graph in the relevant direction with norm $\le C_{\mathrm{AC}}\bar\mu$.

We fix once and for all a geometric constant $C_0 = C_0(N) \ge 1$ absorbing all unit conversions between the half-slab inclusion we verify below and the exact form of the BDV Alt–Caffarelli inequalities.

1.3 Centering hypothesis

Let y_* be the optimal Fraenkel translation. We pre-shift so the comparison ball is B_1. The cost of this translation is recorded in §3 and absorbed into the threshold α_{graph}.

2 From asymmetry to Hausdorff distance

Lemma 2.1 (Symmetric-difference control, [1] L4.2)**.** *$|U\Delta B_1| \le C_\alpha(N)\,\alpha(U)^{1/2}$, with $C_\alpha = C_1^{1/2}$.*

Lemma 2.2 (Density forces Hausdorff smallness)**.** *Set $c_* := \min(c_d, 1/2)$ and*

$$C_H := 4(\omega_N c_*)^{-1/N} + (R+1)(c_* r_d^N)^{-1/N}.$$

Then

$$d_H(\partial U, \partial B_1) \le C_H\,|U\Delta B_1|^{1/N}.$$

Proof. Set $d := d_H(\partial U, \partial B_1)$. Either (Case 1) there is $x \in \partial U$ with $\operatorname{dist}(x, \partial B_1) = d$, or (Case 2) the symmetric statement with roles swapped.

Case 1. $B_{d/2}(x) \cap \partial B_1 = \emptyset$, so $B_{d/2}(x)$ lies in B_1 or in $\mathbb{R}^N \setminus \overline{B_1}$. In the second alternative, density (L4.9) at $x \in \partial U$ at radius $r := \min(d/2, r_d)$ gives $|U \cap B_r(x)| \geq c_d r^N$, and $U \cap B_r(x) \subset U \setminus B_1 \subset U \Delta B_1$, so $c_d r^N \leq |U \Delta B_1|$. Under $d \leq r_d$ this is $c_d (d/2)^N \leq |U \Delta B_1|$. In the first alternative, the complementary density gives $|B_r(x) \setminus U| \geq c_d r^N$ inside $B_1 \setminus U$, same conclusion.

Case 2. $B_{d/2}(\bar{x})$ at $\bar{x} \in \partial B_1$ contains a half-ball of volume $\geq \frac{1}{2}\omega_N (d/2)^N$ inside B_1 or outside B_1; one of these halves lies in $U \Delta B_1$, hence $\frac{1}{2}\omega_N (d/2)^N \leq |U \Delta B_1|$.

If $d > 2r_d$, the same density argument at the fixed radius r_d gives a fixed lower bound for $|U \Delta B_1|$; after increasing the dimensional constant this covers the large-distance alternative too. Combining the small- and large-distance cases gives the stated bound with the chosen C_H. □

Proposition 2.3 (Quantitative asymmetry-to-Hausdorff)**.** *Set* $C'_H(N, R) := C_H(N, R)\, C_\alpha(N)^{1/N}$, $\alpha_0(N, R) := (r_d / C'_H)^{2N}$. *For every selected minimizer* U *with* $\alpha(U) \leq \alpha_0$,

$$d_H(\partial U, \partial B_1) \leq C'_H\, \alpha(U)^{1/(2N)} \leq r_d.$$

Proof. By Lemma 2.1, $|U \Delta B_1| \leq C_\alpha \alpha^{1/2}$; plug into Lemma 2.2. The final inequality follows from $\alpha \leq \alpha_0$. □

3 Barycenter shift is absorbable

Lemma 3.1 (Center shift)**.** *There is* $C'_{\mathrm{Fr}} = C'_{\mathrm{Fr}}(N, R)$ *such that for every selected minimizer* U *with* $\alpha(U) \leq \alpha_0$,

$$|y_*| \leq (R+1)\mathcal{A}(U) \leq (R+1)\sqrt{C_{\mathrm{Fr}}(N)}\, \alpha(U)^{1/2} \leq C'_{\mathrm{Fr}}\, \alpha(U)^{1/2},$$

where $C_{\mathrm{Fr}}(N) := C_1(N)/|B_1|^2$ *from (L4.2).*

Proof. $y_* \in B_{R+1}$ for smallness reasons; the volume identity $|U \Delta (B_1 + y_*)| = |B_1| \mathcal{A}(U)$ combined with $\mathcal{A}^2 \leq C_{\mathrm{Fr}} \alpha$ gives the bound. □

Remark 3.2. The shift $|y_*|$ is dominated by h_F (defined below) when $\alpha \leq (h_F / C'_{\mathrm{Fr}})^2$, absorbed into α_{graph}.

4 From Hausdorff closeness to fixed-scale Alt–Caffarelli flatness

4.1 Choice of flatness scale

With the BDV constants $\bar{\mu}, \bar{\rho}, C_{\mathrm{AC}}$ and the tubular slope threshold $\mu_{\mathrm{tub}}(N) = 1/4$ from §5, define

$$\boxed{\mu := \min\Big(\tfrac{\bar{\mu}}{16 C_0}, \tfrac{\mu_{\mathrm{tub}}}{C_{\mathrm{AC}}}\Big), \quad \rho_\mu := \min\Big(\bar{\rho}, \tfrac{\mu}{18}, \tfrac{r_d}{2}\Big), \quad h_F := \tfrac{\mu \rho_\mu}{16}.}$$

Lemma 4.1 (Geometric slab inclusion)**.** *For* $\bar{x} \in \partial B_1$ *and* $s \leq 6\rho_\mu$,

$$\partial B_1 \cap B_s(\bar{x}) \subset \big\{|(x - \bar{x}) \cdot \nu_{\bar{x}}| \leq s^2/2\big\} \subset \big\{|(x - \bar{x}) \cdot \nu_{\bar{x}}| \leq \mu \rho_\mu\big\}.$$

Proof. B_1 has unit radius, so $|x - \bar{x}| \leq s \Rightarrow |(x - \bar{x}) \cdot \nu_{\bar{x}}| \leq s^2/2$ for $x \in \partial B_1$. For $s \leq 6\rho_\mu$ this is $\leq 18\rho_\mu^2 \leq \mu \rho_\mu$ by $\rho_\mu \leq \mu/18$. □

4.2 Fixed-scale flatness

Proposition 4.2 (Fixed-scale Alt–Caffarelli flatness)**.** *Assume $d_H(\partial U, \partial B_1) \le h_F$. Then for every $\bar{x} \in \partial B_1$ there exists $x_0 \in \partial U$ with $|x_0 - \bar{x}| \le h_F$ such that*

$$\partial U \cap B_{4\rho_\mu}(x_0) \subset \big\{|(x - x_0) \cdot \nu_{\bar{x}}| \le 2\mu\rho_\mu\big\},$$

and the torsion potential u belongs to $F(C_0\mu, 1, +\infty)$ on $B_{4\rho_\mu}(x_0)$ with respect to $-\nu_{\bar{x}}$.

Proof. Step 1. Existence of x_0 is by the Hausdorff bound.

Step 2 (slab inclusion). By Lemma 4.1 at radius $s = 5\rho_\mu + h_F \le 6\rho_\mu$ (since $h_F \le \rho_\mu$ and $\mu \le 1$), $\partial B_1 \cap B_{5\rho_\mu + h_F}(\bar{x})$ lies in $\{|(x - \bar{x}) \cdot \nu_{\bar{x}}| \le \mu\rho_\mu\}$. Every $p \in \partial U \cap B_{5\rho_\mu}(x_0)$ lies within h_F of some $\bar{p} \in \partial B_1 \cap B_{5\rho_\mu + h_F}(\bar{x})$, so $|(p - \bar{x}) \cdot \nu_{\bar{x}}| \le \mu\rho_\mu + h_F$, hence $|(p - x_0) \cdot \nu_{\bar{x}}| \le \mu\rho_\mu + 2h_F \le 2\mu\rho_\mu$ (using $h_F = \mu\rho_\mu/16$).

Step 3 (positive phase, F1). At radius $\rho_\mu \le r_d/2$ the nondegeneracy (L4.12b) gives a point $y \in B_{\rho_\mu}(x_0) \cap \{u > 0\}$ with $u(y) \ge c_d\rho_\mu$. Combined with (L4.12a), the rescaled $u_{x_0,\rho_\mu}(x) := u(x_0 + \rho_\mu x)/\rho_\mu$ satisfies on B_4

$$u_{x_0,\rho_\mu}(x) \ge c_d\big(x \cdot (-\nu_{\bar{x}}) - C_0\mu\big)_+,$$

absorbing constants into C_0. This is (F1).

Step 4 (zero phase, F2). The outer half-ball $\{(x - \bar{x}) \cdot \nu_{\bar{x}} > \mu\rho_\mu\} \cap B_{4\rho_\mu}(\bar{x})$ is disjoint from B_1. By the Hausdorff bound, it is disjoint from U up to a layer of width $h_F < \mu\rho_\mu/16$. Hence $\{(x - x_0) \cdot \nu_{\bar{x}} \ge 3\mu\rho_\mu\} \cap B_{4\rho_\mu}(x_0)$ is contained in U^c, and $u \equiv 0$ there. This is (F2).

Step 5 (Lipschitz upper bound, F3). $\mathrm{Lip}(u) \le L_u$ and the vanishing on the deep zero-layer give

$$u_{x_0,\rho_\mu}(x) \le L_u\big(x \cdot (-\nu_{\bar{x}}) + C_0\mu\big)_+.$$

(F1)+(F2)+(F3) with parameter $C_0\mu \le \bar{\mu}$ (by $\mu \le \bar{\mu}/(16C_0)$) give $u \in F(C_0\mu, 1, +\infty)$ in $B_{4\rho_\mu}(x_0)$ with respect to $-\nu_{\bar{x}}$. □

Remark 4.3 (Orientation check). Step 4 uses the orientation of $\nu_{\bar{x}}$ (outward normal of B_1) to identify the zero-phase side of the slab. The argument is consistent as written: $\mathrm{dist}(\cdot, B_1) > h_F$ on the outer half-ball implies $\mathrm{dist}(\cdot, U) > 0$ by the Hausdorff bound; the connectedness of the deep layer prevents the inner and outer sides of the slab from exchanging roles when $h_F < \mu\rho_\mu/8$.

5 From local flatness to a global $C^{1,\gamma}$ normal graph

5.1 Tubular projection

Lemma 5.1 (Tubular threshold)**.** *Let $\mu_{\mathrm{tub}} := 1/4$. If $g : \partial B_1 \to \mathbb{R}$ satisfies $\|g\|_{C^{0,1}(\partial B_1)} \le \mu_{\mathrm{tub}}$, then $\Phi_g(\theta) := (1 + g(\theta))\theta$ is a bi-Lipschitz embedding, and $|d\Phi_g(\theta)|^2 \ge \frac{15}{36}$.*

Proof. $d\Phi_g = (1 + g)\,\mathrm{id} + dg \otimes \theta$; the bound follows from $(1 + g)^2 \ge (1 - \mu_{\mathrm{tub}})^2 = 9/16$ and $|dg| \le 1/4$. □

5.2 Patching local graphs

Proposition 5.2 (Patching to a global graph)**.** *Assume $d_H(\partial U, \partial B_1) \le h_F$. Then there exists $g : \partial B_1 \to \mathbb{R}$ with*

$$\partial U = \{(1 + g(\theta))\theta : \theta \in \partial B_1\}, \qquad \|g\|_{C^{1,\gamma}(\partial B_1)} \le M_1(N, R) := C_{\mathrm{AC}}\mu \cdot K_{\mathrm{geom}}(N, R),$$

and $\|g\|_{C^{0,1}} \le \mu_{\mathrm{tub}} = 1/4$, where K_{geom} is the finite bi-Lipschitz factor from Lemma 5.1.

Proof. By Proposition 4.2 at every $\bar{x} \in \partial B_1$ there is $x_0(\bar{x})$ such that $u \in F(C_0\mu, 1, +\infty)$ on $B_{4\rho_\mu}(x_0)$ with respect to $-\nu_{\bar{x}}$. Since $C_0\mu \le \bar{\mu}$, BDV Theorem 4.18 applies and $\partial U \cap B_{\rho_\mu}(x_0)$ is a $C^{1,\gamma}$ graph in direction $\nu_{\bar{x}}$ with norm $\le C_{\mathrm{AC}}\mu$.

By $\mu \le \mu_{\mathrm{tub}}/C_{\mathrm{AC}}$, every such local graph has slope $\le \mu_{\mathrm{tub}} = 1/4$. Composing with the inverse of the radial projection of Lemma 5.1 expresses each as a graph $g_{\bar{x}}$ over a neighbourhood of $\bar{x}$ on ∂B_1, with $C^{1,\gamma}$ norm increased by a bounded factor K_{geom}.

Two local graphs $g_{\bar{x}_1}, g_{\bar{x}_2}$ over overlapping caps parameterise the same connected piece of ∂U (by the slope-μ_{tub} tubular non-folding of Lemma 5.1), hence they agree on the overlap, and patch into a global function g with the stated bounds. □

Lemma 5.3 (L^∞ control)**.** *With $C''_H(N,R) := 2C'_H(N,R)$,*

$$\|g\|_{L^\infty(\partial B_1)} \le C''_H\, \alpha(U)^{1/(2N)}.$$

Proof. $|g(\theta)|$ is the radial distance from $(1+g(\theta))\theta \in \partial U$ to ∂B_1; since $\|g\|_{C^{0,1}} \le 1/4$, this is at most 2 times the Euclidean distance, hence $\|g\|_{L^\infty} \le 2d_H(\partial U, \partial B_1) \le 2C'_H\alpha^{1/(2N)} = C''_H\alpha^{1/(2N)}$. □

6 The GraphEntry theorem

Theorem 6.1 (GraphEntry)**.** *There exist $\mu(N,R), \rho_\mu(N,R), h_F(N,R), C'_H(N,R), M_1(N,R)$ and a threshold*

$$\boxed{\alpha_{\mathrm{graph}}(N,R) := \min\{\alpha_0,\ (h_F/C'_H)^{2N},\ (h_F/C'_{\mathrm{Fr}})^2\}}$$

such that the following holds. Let $U \subset B_R$ be a selected minimizer with $|U| = |B_1|$, $\tau \le \tau_{\mathrm{reg}}(N,R)$, and $\alpha(U) \le \alpha_{\mathrm{graph}}(N,R)$. After translating by $-y_*$, ∂U is a global $C^{1,\gamma}$ normal graph over ∂B_1: there exists $g : \partial B_1 \to \mathbb{R}$ with

$$\partial U = \{(1+g(\theta))\theta : \theta \in \partial B_1\},$$

$$\|g\|_{C^{1,\gamma}(\partial B_1)} \le M_1(N,R), \qquad \|g\|_{L^\infty(\partial B_1)} \le C''_H(N,R)\, \alpha(U)^{1/(2N)}.$$

Here $\gamma = \gamma(N,R) \in (0,1)$ is the Alt–Caffarelli exponent of (T4.18).

Proof. By Proposition 2.3, $\alpha(U) \le \alpha_{\mathrm{graph}}$ implies $d_H(\partial U, \partial B_1) \le h_F$ (after threshold reduction absorbing the barycenter shift via Lemma 3.1). Proposition 4.2 then gives fixed-scale Alt–Caffarelli flatness, and Proposition 5.2 + Lemma 5.3 give the global $C^{1,\gamma}$ normal graph with stated bounds. □

Remark 6.2 (Interfaces)*. Input.* The hypothesis that U is selected with $\tau \le \tau_{\mathrm{reg}}$ is the output of SelectionPrinciple; the BDV regularity package is then available with constants (N,R). *Output.* The conclusion is the input of SchauderInterpolation: the uniform $C^{1,\gamma}$ bound M_1 together with the small L^∞ bound $\le C''_H\alpha^{1/(2N)}$ are used there with hodograph + Kinderlehrer–Nirenberg + Schauder bootstrap + interpolation to produce the small C^{2,γ_0} regime of BDV Theorem 3.3. *No compactness.* The proof is contradiction-free.

7 Constant summary

Constant	Origin	Dependence
c_d, r_d	BDV L4.9	N, R
L_u	BDV L4.12a	N, R
$q_\pm, L_q$	BDV L4.16	N, R
C_1, C_2	BDV L4.2	$N,\ R$
$\bar{\mu}, \bar{\rho}, \gamma, C_{\mathrm{AC}}$	BDV T4.18	N, R
$c_* = \min(c_d, 1/2)$	Lem. 2.2	N, R
$C_H = 4(\omega_N c_*)^{-1/N} + (R+1)(c_* r_d^N)^{-1/N}$	Lem. 2.2	N, R
$C'_H = C_H C_\alpha^{1/N}$	Prop. 2.3	N, R
$\alpha_0 = (r_d/C'_H)^{2N}$	Prop. 2.3	N, R
C'_{Fr}	Lem. 3.1	N, R
C_0	Alt–Caffarelli conversion	N
μ, ρ_μ, h_F	§4.1	N, R
$\mu_{\mathrm{tub}} = 1/4$	Lem. 5.1	N
K_{geom}	Prop. 5.2	N, R
$M_1 = C_{\mathrm{AC}}\mu \cdot K_{\mathrm{geom}}$	Prop. 5.2	N, R
$C''_H = 2C'_H$	Lem. 5.3	N, R
$\alpha_{\mathrm{graph}} = \min\{\alpha_0, (h_F/C'_H)^{2N}, (h_F/C'_{\mathrm{Fr}})^2\}$	Thm. 6.1	N, R

References

[1] L. Brasco, G. De Philippis, B. Velichkov, *Faber–Krahn inequalities in sharp quantitative form*, Duke Math. J. **164** (2015), 1777–1831 (arXiv:1306.0392).